



NATIONAL OCCUPATIONAL STANDARD FOR NON- DESTRUCTIVE TEST SPECIALIST

APPROVING AUTHORITY

This National Occupational Standard has been prepared and published under the authority of the Zambia Qualifications Authority Board on 21st September 2021.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards shall be revised every after **5 years**, or whenever necessary, by the issue of either amendments or of revised editions. It is important that users of National Occupational Standards (NOS) should ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Energy National Occupational Standards Development Team, upon which the following organisations were represented:

1. Copperbelt University;
2. Department of Energy - Ministry of Energy;
3. Energy Regulation Board;
4. Kafue Gorge Regional Training Centre;
5. Mines Safety Department - Ministry of Mines and Minerals Development;
6. Ministry of Labour and Social Security Services;
7. Muhanya Solar;
8. Northern Technical College;
9. Rural Electrification Authority;
10. SNV Zambia;
11. University of Zambia;
12. Zambia Qualifications Authority.

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Higher Education established by ZAQA Act No. 13 of 2011 to “***provide for the development and implementation of a national qualifications framework; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing***”.

Among other functions, ZAQA is responsible for “***determining national standards for any occupation***”, through various sector specific National Occupational Standards Development Teams (NOSDTs) of experts composed of representation from appropriate authorities, government departments, industry, academia, regulators, consumer associations and non-governmental organizations, etc.

This National Occupational Standard (NOS) has been developed by the Energy National Occupational Standards Development Team in accordance with the procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula and learning programmes, in various sectors where the occupation exists. In the Energy sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

Non-destructive testing (NDT) is used to inspect or measure the integrity of a material, component/s or structure/s without damaging it for purposes of ensuring safety and quality, as well as to increase production or extend the life of equipment and system components.

NDT is used to test underground pipelines, pipes in industrial plants, reinforcing steel in concrete structures, pressure vessels, wire ropes in suspension bridges and finished machine parts, etc.

NDT services directly contribute to the safety of society. Quality assurance and integrity of critical equipment and facilities that are instrumental in the protection of the environment, public health and life itself. Failure modes result in negative social, financial and environmental implications.

With the foregoing in mind, various sectors in Zambia require non-destructive test Specialists.

This National Occupational Standard highlights core knowledge, skills, competences and personal attributes that Non-destructive test Specialists must possess to be successful in their jobs.

ACRONYMS AND ABBREVIATIONS

CS	Core Skill
NDTS	Non-Destructive Test Specialist
NDT	Non-Destructive Testing
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organizational Knowledge
PC	Performance Criteria
PS	Professional Skill
RK	Regulatory Knowledge
RPL	Recognition of Prior Learning
TK	Technical Knowledge
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Function: is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles, which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

1. OVERVIEW

This is an introductory section providing a brief summary and specific information or commentary about the content of the NOS and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.NDTS.01
Occupation	Non Destructive Testing Engineering
Job Title	Non Destructive Test Specialist
Job Description	A Non-Destructive Test Specialist plans, analyses, develops and tests for conformity to industrial standards. He or She is responsible for adherence to applicable industrial standards and regulations.
Job Purpose	The Non-Destructive Test Specialists work in various industries. These are manufacturing, aerospace, Power Systems, mining, transport and construction among others. The individual on the job conducts tests to ensure safety and quality, increase production or extend the life of equipment, structures and system components.
ZQF Level	Level 7
Sector	Energy
Sub sector	Renewable and Non- Renewable
Other Economic Sector(s) in which the Occupation is Practiced	Oil and Gas, Fabrications, automotive, rubber and plastic, Machined metal products, Construction, Manufacturing, Oil and gas, Buildings, Structures and Machinery, Aerospace, Mining, Academia, Education/Training, Research & Development, regulatory and standards agencies.
Other Similar Jobs that can be performed by a non-destructive test Specialist	Production Engineer, Training/Academia, Design Engineer, Compliance Engineer or Manager.
Minimum Educational Job Entry Qualification(s)	Bachelor's Degree in Engineering or its equivalent
Practicing License Requirements (if any)	Practicing license from Engineers Registration Board (EngRB)
Training/RPL	Prior knowledge in NDT related programs recognized by a recognized accrediting body with minimum 2 years field experience. Aware of Industry Standards, rules, regulations and their applications. Use of ICTs (Hardware and Software necessary for the job) Quality enhancement methods.
Minimum Job Entry Age	21 years
Prior Experience	Minimum of 1 year internship or apprenticeship
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This National Occupational Standard specifies the fundamental knowledge and understanding, skills and competences that a Non-destructive Test Specialist must possess to be successful in his/her job role. It is applicable to Non-destructive Test Specialists working in public, or private organizations or self-employed within or outside the Energy Sector.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

This job requires an individual to possess:

- Creativity;
- Problem solving skills;
- Analytical skills;
- Mathematics skills;
- Integrity and respect for confidentiality;
- Interpersonal skills;
- Commercial Awareness;
- Attention to details;
- Ability to communicate effectively and clearly;
- Self-motivated and team worker;
- Ability to plan and prioritize;
- Quality consciousness; and
- Occupational health and safety oriented.

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into Eleven (11) Units representing the tasks that a job holder should undertake in his/her day to day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This unit is about health, safety and environment].

Unit No.	01
Unit Title	Safety, health and environmental management
Description	This unit is about maintaining safety, health and environmental protection for the individual and the plant
Scope	This unit covers the following: • Health & safety regulations • Environmental protection.
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Health & safety regulations	To be competent, the individual must be able to: PC1.Read, interpret and implement national and organizational safety and health policies and regulations. PC 2. Assess risks and possible safety hazards of all aspects of operations
Environmental protection.	To be competent, the individual must be able to: PC3.Read, interpret and implement the environmental polies for the organisation PC4.Read, interpret and implement environmental standard operating procedures and policies of the organisation PC5. Read, interpret and implement national and global environmental regulations.
Knowledge and Understanding (K)	
A. Organisation -al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1. Company Quality, health and safety policies OK2.Company environmental policies OK3.Company regulations and global best practices
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Safety and health risk assessment TK2. Environmental Risk assessment TK3.Toolbox talk
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Government regulatory agency requirements for health & safety (e.g. Factories act, Mines and Minerals Development Act, Civil Aviation Act, Radiation Protection Act, Petroleum Act, Explosives Act, Engineering Institution of Zambian Act, Environmental Management Act)
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1.write in English and be able to give simple concise instructions in local language.

	Reading Skills
	The individual on the job must be able to: CS2. Read and interpret internal/external documents. CS3. Read and understand manuals, health and safety instructions, memos, other company documents. CS4. Ability to read from different sources- books, screens in machines and signage. CS5. Understand the various colour codes, nomenclature and acronyms related to the profession.
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to : CS6. Express statements or information clearly so that others can hear and understand. CS7. Respond appropriately to any queries relevant to work performance.
B. Professional Skills	Decision Making
	The individual on the job must be able to: PS1. Follow organization rule-based decision-making process. PS2. Take decision with systematic course of actions and/or response.
	Plan and Organise
	The individual on the job must be able to: PS3. Plan and organise work to meet deadlines. PS4. Work constructively and collaboratively with others.
	Customer Centricity
	The individual on the job must be able to: PS5. Follow code of conduct. PS6. Manage relationships with customers with intent on satisfying its requirements for service delivery. .
	Problem Solving and Decision Making
	The individual on the job must be able to: PS7 Recognize problems and search for solutions. PS8. Choose best methods to complete assigned tasks. PS9. Approach relevant authority when required. PS10. Judiciously use common sense in day to day activities
	Analytical Thinking
	The individual on the job must be able to: PS11. Apply domain knowledge, observations and data to select course of action to perform tasks
	Critical Thinking
	The individual on the job must be able to: PS12. Critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities. PS13. Ask relevant questions for better understanding.

UNIT 2 [This unit is about demonstrating competence to adhere to required specifications in the most judicious and economical ways using NDT].

Unit No.	02
Unit Title	Compliance of Structures, Buildings and Machinery to required specifications
Description	This unit is about demonstrating competence to adhere to required specifications in the most judicious and economical ways using NDT.
Scope	This unit covers the following: • Production planning • Product Manufacturing • Quality Management • Assembly of components, subassemblies and final assemblies • Product testing
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Production planning	To be competent, the individual must be able to: PC1: Determine the work (product specifications, output, budget, time) required from the job and discuss the same with the supervisor and subordinates PC2: Plan for production and ensure that schedules, plans, material requirements (using MRP), processes and budget are understood by the production team; PC3: Evaluate and assemble a team of skilled technical staff; PC4: Deploy suitable machinery, equipment and materials PC5: Determine appropriate product/process sketches or blueprints to be used PC6: Examine machine performance and capacity utilisation in order to determine and optimise timelines PC7: Record and report impact of improvement activities PC8: Work safely at all times, complying with health and safety and other relevant regulations and guidelines
Product Manufacturing	To be competent, the individual must be able to: PC9: Lead product manufacturing PC10: Establish product requirements and discuss the same with the supervisor and subordinates; PC11: Determine types of materials, select materials and heat treatment processes and procedures PC12: Manufacture products according to specifications, quantities and timelines PC13: Manage and supervise the production team; PC14: Implement total quality management principles PC15: Comply with health and safety and other relevant regulations and guidelines
Quality Management	To be competent, the individual must be able to: PC16: Interpret product and process specifications; PC17: Deploy Quality Management systems and techniques PC18: Design and implement quality improvement plans and training

	PC19: Collect, document and disseminate data on product quality and improvements to the supervisor and subordinates PC20: Prepare quality improvement reports PC21: Comply with health and safety and other relevant regulations and guidelines
Assembly of components into subassemblies and final assemblies	To be competent, the individual must be able to: PC22: Assemble components into finished products PC23: Test sub and final assemblies PC24: Prepare test reports PC25: Carry product modifications and improvements PC26: Install and commission machines and systems PC27: Comply with health and safety and other relevant regulations and guidelines
Product testing	To be competent, the individual must be able to: PC28: Test and document results PC29: Standardise machine output and calibrate; PC30: Prepare maintenance and operating manuals PC31: Declare products ready for use PC32: Comply with health and safety and other relevant regulations and guidelines
Knowledge and Understanding (K)	
A. Organisation -al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions OK2: Organisation culture and typical customer profile OK3: Company's service level agreements and policies OK4: Company's code of conduct OK5: Organisation pricing and discount policy OK6: Organisation policy on documentation, reporting, etc.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1: Analytical tools and methods TK2: Production tools and methods TK3: Prioritising own and team's workload to ensure that targets are met TK4: Interpreting drawings, charts, specifications, information, data, reports, manuals and other documents needed to understand the requirements of the production task TK5: Keeping up-to-date data and information such as drawings, specifications, manufacturers' manuals and other documents needed in the facility TK6: Operating principles and processes of the manufacturing machines and equipment in use TK7: Using decision making and creativity techniques (such as brainstorming, to generate ideas for improvement) TK8: How to identify training needs and to access training and developmental programmes TK9: Risks and hazards associated with the work, such as handling oils, misuse of tools, using damaged or badly

	maintained tools and equipment, and how to minimise the risks TK10. Prototyping and testing methods and tools TK11. Installation and commissioning techniques TK12. Diagnostic and preventive maintenance techniques TK13: Movement of materials, components or finished goods TK14. Product improvement activities TK15: Engineering safety audits or risk assessments TK16: Quality control/quality assurance
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Applicable national laws regulating the engineering profession RK2: National laws regulating the industry RK3: National Environmental Management Legislation RK4: National Occupational Health and Safety Legislation RK5. Applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and analyse the available data about the products, processes, components, tools and fixtures CS2: Read and interpret sketches and drawings CS3: Read and understand equipment manuals and process documents
	Writing Skills
	The individual on the job must be able to CS4. Note down observations (if any) CS5. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. CS6: Prepare technical and other reports
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS7. Discuss task lists, schedules and activities CS8. Effectively communicate with superiors, colleagues, subordinates and regulators CS9. Attentively listen and comprehend the information given by various sources
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: PS1. Plan and organise the work order and jobs PS2. Organise all process, manuals so that sorting out/accessing information is easy; PS3: Collect and keep up to date production and other information
	Judgment and Critical Thinking
	The individual on the job must be able to: PS4 Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems

	PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to:
	PS7. Follow instructions and work on areas of improvement identified
	PS8. Complete assigned tasks with minimum supervision
	PS9. Complete jobs within timelines and budget and quality norms
	PS10. Be open to other ideas and information
	PS11. Keep up-to-date with latest trends and changes in industry and the profession.
	Problem Solving and Decision Making
	The individual on the job must be able to:
	PS12. Detect problems in day to day tasks
	PS13. Discuss possible solutions to address problems with subordinates and the supervisor
	PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 3 [This Unit is about carrying out installation, testing, modification and commissioning of machines and mechanical systems].

Unit No.	03
Unit Title	Planning and Testing
Description	This unit is about demonstrating competence to carry out planning and testing machines and sub systems
Scope	This unit covers the following: • Planning • Testing
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Planning	To be competent, the individual must be able to: PC1: Prepare bill of material and equipment requirements PC2: Determine number and type of technical staff for the installation PC3: Determine budget and timelines for the job PC4: Assess suitability of location and develop machine layout plan PC5: Prepare and arrange for power (with right voltage) and water PC6: Plan that appropriate hand, power and lifting tools are in place PC7: Plan for adequate ventilation PC8: Complying with health and safety and other relevant regulations and guidelines
Testing	To be competent, the individual must be able to: PC9: Test the installation PC10: Observe and monitor the key performance parameters PC11: Make necessary adjustments or modifications to standardise the machines PC12: Continue testing until pre-set results are obtained PC13: Adherence to operating and maintenance manuals PC14: Complying with health and safety and other relevant regulations and guidelines. PC15: Use visual method of test prior to application of any of the methods of testing.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Site layout TK2: Planning techniques TK3: Testing methods and tools

	TK4. Adherence to operating and maintenance manuals TK5. Maintenance schedules for the machine/component TK6: Selection of suitable fittings for the machine TK7: Preparation of bills of materials and costing TK8: Standards, procedures and guidelines TK9: Providing training to users
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Applicable national laws regulating the engineering profession RK2. National laws regulating the industry; RK3. Environmental Management Act and Legislation RK4. Occupational Health and Safety Act and Legislation RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and analyse the available data about the machines CS2: Read and interpret sketches and drawings CS3: Read equipment manuals and process documents CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: CS5. Note down observations (if any) CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. CS7. Prepare instruction manuals CS8. Prepare technical reports
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS9. Discuss task lists, schedules and activities CS10. Effectively communicate with superiors, colleagues, subordinates and regulators CS11. Listen attentively and comprehend the information given by various sources CS12. Delegate tasks to other staff
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: PS1. Logically plan and organise the work order/schedule and flow of work activities PS2. Organise all processes, manuals so that sorting out/accessing information is easy PS3: Collect and keep up to date records
	Judgment and Critical Thinking
	The individual on the job must be able to: PS4. Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems

	PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to: PS7. Follow instructions and work on areas of improvement identified PS8. Complete assigned tasks with minimum supervision PS9. Complete jobs within timelines and budget and quality norms PS10. Be open to other ideas and information PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Detect problems in day to day tasks PS13. Discuss possible solutions to address problems with subordinates and the supervisor PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 4 [This Unit is about supervision and management of technical staff; artisans, technicians, technologists and other engineers].

Unit No.	04
Unit Title	People management
Description	This unit is about demonstrating competence to supervise and manage other staff
Scope	This unit covers the following: • Skills needs assessment • Job allocation • Supervision and performance management
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Skills needs assessment	To be competent, the individual must be able to: PC1: Comprehensively execute staff induction programme PC2: Carry out a skills audit of staff, PC3: Identify and recommend appropriate training for staff
Job allocation	To be competent, the individual must be able to: PC4: Allocate jobs according to staff abilities PC5: Assemble task teams around highly skilled staff and facilitate team work PC6: Accommodate ideas from both subordinates and superiors.
Supervision and performance management	To be competent, the individual must be able to: PC7: Supervise and manage subordinates; PC8: Provide timely feedback to superiors and subordinates PC9: Carry out activities in the specified sequence and in an agreed timescale PC10. Coach and mentor subordinates PC11. Motivate and provide incentives for outstanding performance.
Knowledge and Understanding (K)	
A. Organisation- al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Delegation TK2: Effective Communication TK3: Motivating Others TK4. Problem Solving Skills TK5. Interpersonal Skills TK6: Conceptual Skills TK7: Technical Skills
C. Regulatory context	The individual on the job must demonstrate knowledge and understanding of:

(Knowledge of Rules and Regulations)	RK1. Applicable national laws regulating the engineering profession RK2. National laws regulating the industry; RK3. Environmental Management Act and Legislation RK4. Occupational Health and Safety Act and Legislation RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and analyse the available data about the machines CS2. Read and interpret sketches and drawings CS3. Read equipment manuals and process documents CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: CS5. Note down observations (if any) CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. CS7. Prepare instruction manuals CS8. Prepare technical reports
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS9. Discuss task lists, schedules and activities CS10. Effectively communicate with superiors, colleagues, subordinates and regulators CS11. Listen attentively and comprehend the information given by various sources CS12. Delegate tasks to other staff
	Plan and Organise
	The individual on the job must be able to: PS1. Logically plan and organise the work order/schedule and flow of work activities PS2. Organise all processes, manuals so that sorting out/accessing information is easy PS3. Collect and keep up to date records
	Judgment and Critical Thinking
	The individual on the job must be able to: PS4. Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to: PS7. Follow instructions and work on areas of improvement identified PS8. Complete assigned tasks with minimum supervision PS9. Complete jobs within timelines and budget and quality norms

	PS10. Be open to other ideas and information PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Detect problems in day to day tasks PS13. Discuss possible solutions to address problems with subordinates and the supervisor PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 5 [This unit is about demonstrating competence to apply the eddy current method of NDT].

Unit No.	05
Unit Title	Eddy Current Testing
Description	This unit is about demonstrating competence to apply the eddy current method of NDT.
Scope	This unit covers the following: • Principle of Operation of Eddy Current techniques • Prototyping and Testing • Product design and procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Principle of operation of Eddy Current Techniques	To be competent, the individual must be able to: PC1: Describe the principle of operation of eddy current, including pulse eddy current non-destructive testing. PC2: State pro, cons and limitations of eddy current testing PC3: Diagnose faults and provide engineering solutions PC4: Prepare test reports on the results with recommendations. PC5: Conduct calibration of eddy current instruments PC6: Conduct Non-destructive testing using conventional and or pulse eddy current.
Prototyping and testing	To be competent, the individual must be able to: PC7: Develop and test new products and prototypes PC8: Make the required modifications to the product according to the test results PC9: Prepare a test report on the results with recommendations for the manufacture of the tested product PC10: Work safely at all times, complying with health and safety and other relevant regulations and guidelines PC11: Acquaint him or herself with data acquisition software
Product design and procedures	To be competent, the individual must be able to: PC12: Recommend improvements to existing or new equipment and products PC13: Monitor machine performance and capacity utilisation in order to determine and optimise functions PC14: Work safely at all times, complying with health and safety and other relevant regulations and guidelines
Knowledge and Understanding (K)	
A. Organisation- al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company

B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Site layout - TK2. Planning techniques - TK3. Testing methods and tools - TK4. Adherence to operating and maintenance manuals - TK5. Maintenance schedules for the machine/component - TK6. Selection of suitable fittings for the machine - TK7. Preparation of bills of materials and costing - TK8. Standards, procedures and guidelines - TK9. Providing training to users
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: - RK1. Applicable national laws regulating the engineering profession - RK2. National laws regulating the industry; - RK3. Environmental Management Act and Legislation - RK4. Occupational Health and Safety Act and Legislation - RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: - CS1. Read and analyse the available data about the machines - CS2. Read and interpret sketches and drawings - CS3. Read equipment manuals and process documents - CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: - CS5. Note down observations (if any) - CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. - CS7. Prepare instruction manuals - CS8. Prepare technical reports
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: - CS9. Discuss task lists, schedules and activities - CS10. Effectively communicate with superiors, colleagues, subordinates and regulators - CS11. Listen attentively and comprehend the information given by various sources - CS12. Delegate tasks to other staff
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: - PS1. Logically plan and organise the work order/schedule and flow of work activities - PS2. Organise all processes, manuals so that sorting out/accessing information is easy - PS3. Collect and keep up to date records
	Judgment and Critical Thinking
	The individual on the job must be able to:

	PS4. Use common sense and make judgments in day to day activities
	PS5. Use reasoning skills to identify and resolve basic problems
	PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to: PS7. Follow instructions and work on areas of improvement identified PS8. Complete assigned tasks with minimum supervision PS9. Complete jobs within timelines and budget and quality norms PS10. Be open to other ideas and information PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Detect problems in day to day tasks PS13. Discuss possible solutions to address problems with subordinates and the supervisor PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 6 [This unit is about demonstrating competence to apply the Liquid penetrant method of NDT].

Unit No.	06
Unit Title	Liquid Penetrant NDT
Description	This unit is about demonstrating competence to apply the Liquid penetrant method of NDT.
Scope	This unit covers the following: • Principle of Operation of Liquid penetrant techniques • Prototyping and Testing • Product design and procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Principle of operation of Liquid penetrant techniques	To be competent, the individual must be able to: PC1: Describe the principle of operation of Liquid penetrant non-destructive testing. PC2: State pros, cons and limitations of liquid penetrant testing PC3: Diagnose faults and provide engineering solutions PC4: Prepare test reports on the results with recommendations. PC5: Conduct calibration of liquid penetrant instruments PC6: Conduct Non-destructive testing using liquid penetrant
Prototyping, and testing	To be competent, the individual must be able to: PC7: Develop and test new products and prototypes PC8: Make the required modifications to the product according to the test results PC9: Prepare a test report on the results with recommendations for the manufacture of the tested product PC10: Work safely at all times, complying with health and safety and other relevant regulations and guidelines PC11: Acquaint him or herself with data acquisition software
Product design and procedures	To be competent, the individual must be able to: PC12: Lead on making improvements to products and procedures PC13: Design suitable products using technical data and other information PC14: Utilise customer requirements and the design specification for new product and processes PC15: Recommend improvements to existing or new equipment and products PC16: Monitor machine performance and capacity utilisation in order to determine and optimise functions PC17: Work safely at all times, complying with health and safety and other relevant regulations and guidelines
Knowledge and Understanding (K)	
A. Organisation- al Context (Knowledge of the company/ organisation)	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company

and its processes)	
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Site layout - TK2. Planning techniques - TK3. Testing methods and tools - TK4. Adherence to operating and maintenance manuals - TK5. Maintenance schedules for the machine/component - TK6. Selection of suitable fittings for the machine - TK7. Preparation of bills of materials and costing - TK8. Standards, procedures and guidelines - TK9. Providing training to users
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: - RK1. Applicable national laws regulating the engineering profession - RK2. National laws regulating the industry; - RK3. Environmental Management Act and Legislation - RK4. Occupational Health and Safety Act and Legislation - RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: - CS1. Read and analyse the available data about the machines - CS2. Read and interpret sketches and drawings - CS3. Read equipment manuals and process documents - CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: - CS5. Note down observations (if any) - CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. - CS7. Prepare instruction manuals - CS8. Prepare technical reports
B. Professional Skills	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: - CS9. Discuss task lists, schedules and activities - CS10. Effectively communicate with superiors, colleagues, subordinates and regulators - CS11. Listen attentively and comprehend the information given by various sources - CS12. Delegate tasks to other staff
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: - PS1. Logically plan and organise the work order/schedule and flow of work activities - PS2. Organise all processes, manuals so that sorting out/accessing information is easy - PS3. Collect and keep up to date records

	Judgment and Critical Thinking
	The individual on the job must be able to: PS4. Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to: PS7. Follow instructions and work on areas of improvement identified PS8. Complete assigned tasks with minimum supervision PS9. Complete jobs within timelines and budget and quality norms PS10. Be open to other ideas and information PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Detect problems in day to day tasks PS13. Discuss possible solutions to address problems with subordinates and the supervisor PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 7 [This unit is about demonstrating competence to apply the Magnetic particle method of NDT].

Unit No.	07
Unit Title	Magnetic Particle NDT
Description	This unit is about demonstrating competence to apply the Magnetic Particle method of NDT.
Scope	This unit covers the following: • Principle of Operation of Magnetic Particle techniques • Prototyping and Testing • Product design and procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Principle of operation of Magnetic Particle Techniques	To be competent, the individual must be able to: PC1: Apply the principle of operation of Magnetic particle non-destructive testing. PC2: State pro, cons and limitations of liquid magnetic particle testing PC3: Diagnose faults and provide engineering solutions PC4: Prepare test reports on the results with recommendations. PC5: Conduct calibration of magnetic particle instruments PC6: Conduct Non-destructive testing using magnetic particle
Prototyping, and testing	To be competent, the individual must be able to: PC7: Develop and test new products and prototypes PC8: Make the required modifications to the product according to the test results PC9: Prepare a test report on the results with recommendations for the manufacture of the tested product PC10: Work safely at all times, complying with health and safety and other relevant regulations and guidelines PC11: Acquaint him or herself with data acquisition software
Product design and procedures	To be competent, the individual must be able to: PC12: Lead on making improvements to products and procedures PC13: Design suitable products using technical data and other information PC14: Utilise customer requirements and the design specification for new product and processes PC15: Recommend improvements to existing or new equipment and products PC16: Monitor machine performance and capacity utilisation in order to determine and optimise functions PC17: Work safely at all times, complying with health and safety and other relevant regulations and guidelines
Knowledge and Understanding (K)	
A. Organisation- al Context (Knowledge of the company/	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company

organisation and its processes)	
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Site layout TK2. Planning techniques TK3. Testing methods and tools TK4. Adherence to operating and maintenance manuals TK5. Maintenance schedules for the machine/component TK6. Selection of suitable fittings for the machine TK7. Preparation of bills of materials and costing TK8. Standards, procedures and guidelines TK9. Providing training to users
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Applicable national laws regulating the engineering profession RK2. National laws regulating the industry; RK3. Environmental Management Act and Legislation RK4. Occupational Health and Safety Act and Legislation RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and analyse the available data about the machines CS2. Read and interpret sketches and drawings CS3. Read equipment manuals and process documents CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: CS5. Note down observations (if any) CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. CS7. Prepare instruction manuals CS8. Prepare technical reports
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS9. Discuss task lists, schedules and activities CS10. Effectively communicate with superiors, colleagues, subordinates and regulators CS11. Listen attentively and comprehend the information given by various sources CS12. Delegate tasks to other staff
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: PS1. Logically plan and organise the work order/schedule and flow of work activities

	PS2. Organise all processes, manuals so that sorting out/accessing information is easy PS3: Collect and keep up to date records
	Judgment and Critical Thinking
	The individual on the job must be able to: PS4. Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to: PS7. Follow instructions and work on areas of improvement identified PS8. Complete assigned tasks with minimum supervision PS9. Complete jobs within timelines and budget and quality norms PS10. Be open to other ideas and information PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Detect problems in day to day tasks PS13. Discuss possible solutions to address problems with subordinates and the supervisor PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 8 [This unit is about demonstrating competence to apply the Ultrasonic method of NDT].

Unit No.	08
Unit Title	Ultrasonic NDT
Description	This unit is about demonstrating competence to apply the Ultrasonic method of NDT.
Scope	This unit covers the following: • Principle of Operation of Ultrasonic techniques • Prototyping and Testing • Product design and procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Principle of operation of Ultrasonic Techniques	To be competent, the individual must be able to: PC1: Describe the principle of operation of Ultrasonic non-destructive testing. PC2: State pro, cons and limitations of Ultrasonic testing PC3: Diagnose faults and provide engineering solutions PC4: Prepare test reports on the results with recommendations. PC5: Conduct calibration of Ultrasonic instruments PC6: Conduct Non-destructive testing using magnetic particle
Prototyping, and testing	To be competent, the individual must be able to: PC7: Develop and test new products and prototypes PC8: Make the required modifications to the product according to the test results PC9: Prepare a test report on the results with recommendations for the manufacture of the tested product PC10: Work safely at all times, complying with health and safety and other relevant regulations and guidelines PC11: Acquaint him or herself with data acquisition software
Product design and procedures	To be competent, the individual must be able to: PC12: Lead on making improvements to products and procedures PC13: Design suitable products using technical data and other information PC14: Utilise customer requirements and the design specification for new product and processes PC15: Recommend improvements to existing or new equipment and products PC16: Monitor machine performance and capacity utilisation in order to determine and optimise functions PC17: Work safely at all times, complying with health and safety and other relevant regulations and guidelines
Knowledge and Understanding (K)	
A. Organisation- al Context (Knowledge of the company/ organisation)	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company

and its processes)	
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Site layout - TK2. Planning techniques - TK3. Testing methods and tools - TK4. Adherence to operating and maintenance manuals - TK5. Maintenance schedules for the machine/component - TK6. Selection of suitable fittings for the machine - TK7. Preparation of bills of materials and costing - TK8. Standards, procedures and guidelines - TK9. Providing training to users
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: - RK1. Applicable national laws regulating the engineering profession - RK2. National laws regulating the industry; - RK3. Environmental Management Act and Legislation - RK4. Occupational Health and Safety Act and Legislation - RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: - CS1. Read and analyse the available data about the machines - CS2. Read and interpret sketches and drawings - CS3. Read equipment manuals and process documents - CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: - CS5. Note down observations (if any) - CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. - CS7. Prepare instruction manuals - CS8. Prepare technical reports
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: - CS9. Discuss task lists, schedules and activities - CS10. Effectively communicate with superiors, colleagues, subordinates and regulators - CS11. Listen attentively and comprehend the information given by various sources - CS12. Delegate tasks to other staff
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: - PS1. Logically plan and organise the work order/schedule and flow of work activities - PS2. Organise all processes, manuals so that sorting out/accessing information is easy

	PS3: Collect and keep up to date records
	Judgment and Critical Thinking
	The individual on the job must be able to: PS4. Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to: PS7. Follow instructions and work on areas of improvement identified PS8. Complete assigned tasks with minimum supervision PS9. Complete jobs within timelines and budget and quality norms PS10. Be open to other ideas and information PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Detect problems in day to day tasks PS13. Discuss possible solutions to address problems with subordinates and the supervisor PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 9 [This unit is about demonstrating competence to apply the Radiographic method of NDT].

Unit No.	09
Unit Title	Radiographic NDT
Description	This unit is about demonstrating competence to apply the Radiographic method of NDT.
Scope	This unit covers the following: • Principle of Operation of Radiographic techniques • Prototyping and Testing • Product design and procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Principle of operation of Radiography Techniques	To be competent, the individual must be able to: PC1: Describe the principle of operation of Radiographic non-destructive testing. PC2: State pro, cons and limitations of Ultrasonic testing PC3: Diagnose faults and provide engineering solutions PC4: Prepare test reports on the results with recommendations. PC5: Conduct calibration of Radiographic instruments PC6: Conduct Non-destructive testing using Radiographic techniques
Prototyping, and testing	To be competent, the individual must be able to: PC7: Develop and test new products and prototypes PC8: Make the required modifications to the product according to the test results PC9: Prepare a test report on the results with recommendations for the manufacture of the tested product PC10: Work safely at all times, complying with health and safety and other relevant regulations and guidelines PC11: Acquaint him or herself with data acquisition software
Product design and procedures	To be competent, the individual must be able to: PC12: Lead on making improvements to products and procedures PC13: Design suitable products using technical data and other information PC14: Utilise customer requirements and the design specification for new product and processes PC15: Recommend improvements to existing or new equipment and products PC16: Monitor machine performance and capacity utilisation in order to determine and optimise functions PC17: Work safely at all times, complying with health and safety and other relevant regulations and guidelines
Knowledge and Understanding (K)	
A. Organisation- al Context (Knowledge of the company/ organisation)	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company

and its processes)	
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: - TK1. Site layout - TK2. Planning techniques - TK3. Testing methods and tools - TK4. Adherence to operating and maintenance manuals - TK5. Maintenance schedules for the machine/component - TK6. Selection of suitable fittings for the machine - TK7. Preparation of bills of materials and costing - TK8. Standards, procedures and guidelines - TK9. Providing training to users
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: - RK1. Applicable national laws regulating the engineering profession - RK2. National laws regulating the industry; - RK3. Environmental Management Act and Legislation - RK4. Occupational Health and Safety Act and Legislation - RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: - CS1. Read and analyse the available data about the machines - CS2. Read and interpret sketches and drawings - CS3. Read equipment manuals and process documents - CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: - CS5. Note down observations (if any) - CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. - CS7. Prepare instruction manuals - CS8. Prepare technical reports
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: - CS9. Discuss task lists, schedules and activities - CS10. Effectively communicate with superiors, colleagues, subordinates and regulators - CS11. Listen attentively and comprehend the information given by various sources - CS12. Delegate tasks to other staff
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: - PS1. Logically plan and organise the work order/schedule and flow of work activities - PS2. Organise all processes, manuals so that sorting out/accessing information is easy

	PS3: Collect and keep up to date records
	Judgment and Critical Thinking
	The individual on the job must be able to: PS4. Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to: PS7. Follow instructions and work on areas of improvement identified PS8. Complete assigned tasks with minimum supervision PS9. Complete jobs within timelines and budget and quality norms PS10. Be open to other ideas and information PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Detect problems in day to day tasks PS13. Discuss possible solutions to address problems with subordinates and the supervisor PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 10 [This unit is about demonstrating competence to apply the Infrared thermographic method of NDT].

Unit No.	10
Unit Title	Infrared Thermographic NDT
Description	This unit is about demonstrating competence to apply the Infrared thermographic method of NDT.
Scope	This unit covers the following: • Principle of Operation of Infrared thermographic testing • Prototyping and Testing • Product design and procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Principle of operation of Infrared Techniques	To be competent, the individual must be able to: PC1: Describe the principle of operation of Infrared thermographic non-destructive testing. PC2: State pro, cons and limitations of Infrared thermographic testing PC3: Diagnose faults and provide engineering solutions PC4: Prepare test reports on the results with recommendations. PC5: Conduct calibration of Infrared thermographic instruments PC6: Conduct Non-destructive testing using Infrared thermographic techniques
Prototyping, and testing	To be competent, the individual must be able to: PC7: Develop and test new products and prototypes PC8: Make the required modifications to the product according to the test results PC9: Prepare a test report on the results with recommendations for the manufacture of the tested product PC10: Work safely at all times, complying with health and safety and other relevant regulations and guidelines PC11: Acquaint him or herself with data acquisition software
Product design and procedures	To be competent, the individual must be able to: PC12: Lead on making improvements to products and procedures PC13: Design suitable products using technical data and other information PC14: Utilise customer requirements and the design specification for new product and processes PC15: Recommend improvements to existing or new equipment and products PC16: Monitor machine performance and capacity utilisation in order to determine and optimise functions PC17: Work safely at all times, complying with health and safety and other relevant regulations and guidelines
Knowledge and Understanding (K)	
A. Organisation- al Context (Knowledge of	The individual on the job must demonstrate knowledge and understanding of:

the company/organisation and its processes)	OK1: Legislation, standards, policies, and procedures followed in the company
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Site layout TK2: Planning techniques TK3: Testing methods and tools TK4. Adherence to operating and maintenance manuals TK5. Maintenance schedules for the machine/component TK6: Selection of suitable fittings for the machine TK7: Preparation of bills of materials and costing TK8: Standards, procedures and guidelines TK9: Providing training to users
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Applicable national laws regulating the engineering profession RK2. National laws regulating the industry; RK3. Environmental Management Act and Legislation RK4. Occupational Health and Safety Act and Legislation RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and analyse the available data about the machines CS2: Read and interpret sketches and drawings CS3: Read equipment manuals and process documents CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: CS5. Note down observations (if any) CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. CS7. Prepare instruction manuals CS8. Prepare technical reports
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS9. Discuss task lists, schedules and activities CS10. Effectively communicate with superiors, colleagues, subordinates and regulators CS11. Listen attentively and comprehend the information given by various sources CS12. Delegate tasks to other staff
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: PS1. Logically plan and organise the work order/schedule and flow of work activities

	PS2. Organise all processes, manuals so that sorting out/accessing information is easy PS3: Collect and keep up to date records
	Judgment and Critical Thinking
	The individual on the job must be able to: PS4. Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to: PS7. Follow instructions and work on areas of improvement identified PS8. Complete assigned tasks with minimum supervision PS9. Complete jobs within timelines and budget and quality norms PS10. Be open to other ideas and information PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to: PS12. Detect problems in day to day tasks PS13. Discuss possible solutions to address problems with subordinates and the supervisor PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

UNIT 11 [This unit is about demonstrating competence to apply Visual Testing method of NDT].

Unit No.	11
Unit Title	Visual Testing NDT
Description	This unit is about demonstrating competence to apply the Visual Testing method of NDT.
Scope	This unit covers the following: • Principle of Visual Inspection Testing • Prototyping and Testing • Product design and procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria (PC)
Principle of Visual Inspection Testing	To be competent, the individual must be able to: PC1: Describe Visual Inspection (VI) testing method PC2: State pro, cons and limitations of visual inspection testing. PC3: Inspect for flaws that are visible to the naked eye to allow for a feasible and fast control of quality at every step of the fabrication or maintenance process.
Prototyping, and Testing	To be competent, the individual must be able to: PC4: Develop and test new products and prototypes PC5: Make the required modifications to the product according to the test results PC6: Prepare a test report on the results with recommendations for the manufacture of the tested product PC7: Work safely at all times, complying with health and safety and other relevant regulations and guidelines PC8: Acquaint him or herself with data acquisition software
Product design and procedures	To be competent, the individual must be able to: PC9: Lead on making improvements to products and procedures PC10: Design suitable products using technical data and other information PC11: Utilise customer requirements and the design specification for new product and processes PC12: Recommend improvements to existing or new equipment and products PC13: Monitor machine performance and capacity utilisation in order to determine and optimise functions PC14: Work safely at all times, complying with health and safety and other relevant regulations and guidelines
Knowledge and Understanding (K)	
A. Organisation- al Context (Knowledge of the company/ organisation and its processes)	The individual on the job must demonstrate knowledge and understanding of: OK1: Legislation, standards, policies, and procedures followed in the company
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of:

	TK1. Site layout TK2: Planning techniques TK3: Testing methods and tools TK4. Adherence to operating and maintenance manuals TK5. Maintenance schedules for the machine/component TK6: Selection of suitable fittings for the machine TK7: Preparation of bills of materials and costing TK8: Standards, procedures and guidelines TK9: Providing training to users
C. Regulatory context (Knowledge of Rules and Regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Applicable national laws regulating the engineering profession RK2. National laws regulating the industry; RK3. Environmental Management Act and Legislation RK4. Occupational Health and Safety Act and Legislation RK5. Other applicable National and International Standards
Skills (S)	
A. Core Skills/ Generic Skills	Reading Skills
	The individual on the job must be able to: CS1. Read and analyse the available data about the machines CS2: Read and interpret sketches and drawings CS3: Read equipment manuals and process documents CS4. Read and interpret internal information sent by supervisor/other teams
	Writing Skills
	The individual on the job must be able to: CS5. Note down observations (if any) CS6. Prepare requisitions to procurement/stores on the requirement of materials, tools and equipment, etc. CS7. Prepare instruction manuals CS8. Prepare technical reports
	Oral Communication (Listening and Speaking skills)
	The individual on the job must be able to: CS9. Discuss task lists, schedules and activities CS10. Effectively communicate with superiors, colleagues, subordinates and regulators CS11. Listen attentively and comprehend the information given by various sources CS12. Delegate tasks to other staff
B. Professional Skills	Plan and Organise
	The individual on the job must be able to: PS1. Logically plan and organise the work order/schedule and flow of work activities PS2. Organise all processes, manuals so that sorting out/accessing information is easy PS3: Collect and keep up to date records
	Judgment and Critical Thinking
	The individual on the job must be able to: PS4. Use common sense and make judgments in day to day activities PS5. Use reasoning skills to identify and resolve basic problems

	PS6. Use intuition to detect any potential problems which could arise during operations
	Desire to Learn and Take Initiatives
	The individual on the job must be able to:
	PS7. Follow instructions and work on areas of improvement identified
	PS8. Complete assigned tasks with minimum supervision
	PS9. Complete jobs within timelines and budget and quality norms
	PS10. Be open to other ideas and information
	PS11. Keep up-to-date with latest trends and changes in industry and the profession
	Problem Solving and Decision Making
	The individual on the job must be able to:
	PS12. Detect problems in day to day tasks
	PS13. Discuss possible solutions to address problems with subordinates and the supervisor
	PS14. Make decisions in emergency situations in the absence of the supervisor (as per the authority matrix defined by the organisation)

5. EQUIPMENT, TOOLS AND CONSUMABLE MATERIALS

These include, but not limited to; Computer and Printer, Personal protective equipment (PPE), Testing equipment and Testing materials

6. DILEMMAS/CHALLENGES AND COMPLEXITIES FOR A JOB HOLDER

Dilemmas associated with the job of an NDT engineer include long working hours, exposure to chemical, physical and biological hazards, time pressure to complete tasks, working in extreme weather such as hot and cold conditions, working in noisy, wet and dusty environments, etc.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to dilemmas include carrying out risk assessment and implementing appropriate control measures, ensuring good time management and planning, participating in workplace safety sensitization and awareness, supporting capacity building through training, managing work stress, adhering to company's safety and standard operating procedures at all times, paying attention to detail, consulting extensively within and outside one's department/team on safety and other issues.

7. WORKING CONDITIONS/ENVIRONMENT

Working conditions include working in cold, hot and wet conditions, working at heights, stand/walk for long hours, working in laboratory environment, working in shifts, areas that are noisy and dusty, areas with limited lighting and ventilation. Working in confined spaces, working in explosives environment.

8. PARTIES INVOLVED/INTERACTING WITH THE JOB HOLDER OR TRAINEE

8.1 Internal/Within the Organization

Parties involved/interacting with the job holder who are internal to the organization include supervisors, subordinates, and other employees (e.g. architects, surveyors, engineers: civil, building, electrical and mechanical), trainers, safety team, etc.

8.2 External/Outside the Organization

Parties involved/interacting with the job holder who are external include customers/clients, government regulators, trainers, suppliers of equipment/tools/consumables, Engineers from other companies, labour unions/occupational health and safety associations, Academia etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions;
- Be able to walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Be able to lift relatively heavy materials, tools and equipment;
- Be able to use fingers, hands and feet with ease to complete the assigned task (dexterity).

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

- A.1.1** Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programmes developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programmes developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

- A.1.2** Awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.NDTS.01		
ZQF Level	Level 7	Version Number	01
Sector	Energy	Date of Approval	September 2021
Sub Sector	Buildings, Structures and Machinery.	Date of Last Review	N/A
Occupation	Non Destructive Test Specialist	Date of Next Review	September 2026

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